路政署
Highways Department

Agreement No. CE 4/2024 (HY)

Northern Metropolis Highway – Investigation

Environmental Impact Assessment Report - Executive Summary (San Tin Section)

March 2026

Prepared and checked by:



16 March 2026

Wong Chi Wah
HKIQEP EIA Expert
(Membership No. PM0075)

AECOM-ATKINSREALIS JV

Disclaimer:

*The information contained in this document originally produced by AAJV was produced solely for the use of the **Highways Department** and was prepared to assist in **Agreement No. CE 4/2024 (HY) Northern Metropolis Highway – Investigation**. The information contained in this document is not intended to be used by **Highways Department** in connection with any project financing or fund raising purposes and AAJV does not accept any duty to, nor accepts any responsibility to, any other party who may rely upon such information for such purposes. AAJV devoted normal professional efforts compatible with the time and budget available in the bid process. AAJV's findings represent its reasonable judgments within the time and budget context of its commission and utilizing the information available to it at the time. Neither AAJV nor its parent corporation, or its affiliates, (a) makes any warranty, expressed or implied, with respect to the use of any information or methods disclosed in this document or (b) assumes any liability with respect to the use of any information or methods disclosed in this document. Any recipient of this document, by their acceptance or use of this document, releases AAJV , its parent corporation, and its and their affiliates from any liability for direct, indirect, consequential or special loss or damage whether arising in contract, warranty, express or implied, tort or otherwise, and irrespective of fault, negligence and strict liability. AAJV undertakes no duty to, nor accepts any responsibility to, any other party who may rely upon such information unless otherwise agreed or consented to by AAJV in writing (including, without limitation, in the form of a reliance letter) herein or in a separate document. Any party who is entitled to rely on this document may do so only on the document in its entirety and not on any excerpt or summary. Entitlement to rely upon this document is conditional upon the entitled party accepting full responsibility and not holding AAJV liable in any way for any impacts on its work product for Project name arising from changes in "external" factors such as changes in government policy, in the pricing of goods and materials or changes in the owner's policy affecting the operation of the project. This document may include "forward-looking statements". These statements relate to AAJV's expectations, beliefs, intentions or strategies regarding the future. These statements may be identified by the use of words like "anticipate," "believe," "estimate," "expect," "intend," "may," "plan," "project," "will," "should," "seek," and similar expressions. The forward-looking statements reflect AAJV's views and assumptions with respect to future events as of the date of this report and are subject to future economic conditions, and other risks and uncertainties. Actual and future results and trends could differ materially from those set forth in such statements due to various factors, including, without limitation, those discussed in this report. These factors are beyond AAJV's ability to control or predict. No section or element of this document produced by AAJV may be removed from this document, reproduced, electronically stored or transmitted in any form by parties other than those for whom the document has been prepared without the written permission of AAJV.*

TABLE OF CONTENTS

Page

1. INTRODUCTION 1
1.1 Project Title 1
1.2 Background 1
1.3 Environmental Impact Assessment (EIA) Study 2
1.4 Purpose of this Executive Summary 3
2. PROJECT DESCRIPTION 4
2.1 Objective and Scope of the Project 4
2.2 Designated Project under the EIAO 4
2.3 Need of the Project 5
2.4 Benefits of the Project 5
2.5 Consideration of “With” and “Without” Project Scenarios 9
2.6 Consideration of Alignment Options 10
2.7 Construction Methodologies for Key Design Elements 14
2.8 Implementation Programme 15
3. KEY FINDINGS OF THE ENVIRONMENTAL IMPACT ASSESSMENT 17
3.1 Approach to the EIA 17
3.2 Air Quality Impact 18
3.3 Noise Impact 19
3.4 Water Quality Impact 20
3.5 Waste Management Implications 21
3.6 Land Contamination 22
3.7 Ecology 23
3.8 Fisheries 24
3.9 Landscape and Visual Impact 25
3.10 Cultural Heritage 26
3.11 Hazard to Life 27
4. ENVIRONMENTAL MONITORING AND AUDIT (EM&A) 27
5. SUMMARY OF ENVIRONMENTAL OUTCOMES 27
6. CONCLUSION 29

List of Tables

Table 2.1	Schedule 2 Designated Projects in this Project
Table 2.2	Summary of Alternative Mitigation Measures
Table 2.3	Alignment Options Comparison
Table 2.4	Summary of Possible Construction Methods for the Project
Table 2.5	Tentative Construction Programme
Table 5.1	Summary of Key Environmental Problems Avoided and Sensitive Areas Protected
Table 6.1	Summary of Environmental Impacts

List of Figures

R40/Figure 1.1	Location of the Project – San Tin Section
R40/Figure 1.2	Location of Designated Project – San Tin Section
R40/Figure 1.3	Alignment Options
R40/Figure 1.3.1	Alignment Options (Sheet 1 of 4)
R40/Figure 1.3.2	Alignment Options (Sheet 2 of 4)
R40/Figure 1.3.3	Alignment Options (Sheet 3 of 4)
R40/Figure 1.3.4	Alignment Options (Sheet 4 of 4)

1. INTRODUCTION

1.1 Project Title

1.1.1.1 The title of the Project is “Northern Metropolis Highway – San Tin Section” (hereafter referred to as the Project).

1.1.1.2 The Project is one of the four sections of the Northern Metropolis Highway (NMH), with the other three being the Tin Shui Wai Section, Kwu Tung Section and New Territories North (NTN) New Town Section. The NMH has an overall length of approximately 24 km.

1.2 Background

1.2.1.1 In December 2020, the Transport Department (TD) commissioned a consultancy study titled “Strategic Study on Major Roads beyond 2030 – Feasibility Study” (SSMR FS) under Agreement No. CE 7/2018 (TT). The SSMR FS examined the future configuration of Hong Kong’s major road infrastructure to ensure that the planning of major transport infrastructure will drive or even reserve capacity to meet Hong Kong's long-term development needs, including those proposed in the Northern Metropolis Development Strategy.

1.2.1.2 The 2022 Policy Address mentioned that three major road projects recommended under the Strategy Study on Major Roads beyond 2030 (SSMR2030+) would be taken forward to create a highly interconnected and accessible road network, significantly driving and supporting Hong Kong's future development. NMH is one of the three major road projects.

1.2.1.3 The 2023 Policy Address announced an extension of the NMH proposal to the east, connecting it to the Sha Tau Kok Road Interchange.

1.2.1.4 The NMH aims to establish a new strategic route improving east-west connectivity within the Northern Metropolis and reduce potential traffic congestion on Yuen Long Highway, San Tin Highway, and Fanling Highway.

1.2.1.5 The NMH is being taken forward following the principle of “Holistic Planning and Phased Implementation”. The investigation study for the entire NMH will be completed first, followed by detailed design and construction of individual sections to cater for the development needs and suit the allocation of public resources. The San Tin Section is tentatively ready for tendering in Year 2027. Subject to the further planning and the actual implementation of the Project, advance works and the associated study would tentatively commence in the same year or Year 2028 with target commissioning of the Project in or before 2036.

1.2.1.6 The NMH will take into account the transport infrastructure projects in the pipeline, including the capacity of roads and railways, overall cost-effectiveness and allocation of public resources, etc., in formulating a more concrete implementation programme, with a view to completing the remaining sections of the NMH in phases in about 3 to 4 years after the commissioning of the San Tin Section.

1.2.1.7 Being a part of the Northern Metropolis Highway, the Project comprises mainly a dual three-lane carriageway in the form of at-grade road, tunnel and viaduct, starts from the interchange of Yuen Long Highway / Tsing Long Highway on the west and

connects to Fanling Highway (Pak Shek Au Section), with associated interchanges and ancillary facilities.

1.2.1.8 AECOM Asia Company Limited and AtkinsRealis Asia Limited Joint Venture has been commissioned by the Highways Department to undertake Agreement No. CE 4/2024 (HY) – "Northern Metropolis Highway – Investigation", which commenced in March 2025. During the investigation, the Project has incorporated environmental benefit elements and integrated ecological-friendly design measures. For details, please refer to **Section 2.4.**

1.3 Environmental Impact Assessment (EIA) Study

1.3.1.1 A Project Profile (PP) (No. PP-683/2025) for the Project based on the 8.5-kilometer-long alignment in the earlier stage of the Study was submitted to the Environmental Protection Department (EPD) on 22 April 2025 for the application for an EIA Study Brief under Section 5(1)(a) of the *Environmental Impact Assessment Ordinance* (EIAO). An EIA Study Brief No. ESB-373/2025 was subsequently issued on 5 June 2025.

1.3.1.2 Subsequent to the issue of the EIA Study Brief (No. ESB-373/2025), some changes were made to the project boundary, layout and design to facilitate the project development. In accordance with Clause 6.2 of the EIA Study Brief, confirmation had been sought from the Director of Environmental Protection in writing to confirm that the scope of issues covered by the EIA Study Brief (No. ESB-373/2025) can still cover the key changes, and the additional issues, if any, that the EIA report must address. Having reviewed the proposed changes, EPD confirmed that the EIA Study Brief is still valid.

1.3.1.3 The EIA study has been conducted for the Project in accordance with the requirements of the EIA Study Brief (No. ESB-373/2025) and the *Technical* Memorandum *on Environmental Impact Assessment Process* (EIAO-TM). The purposes of this EIA study are to identify information on the nature and extent of potential environmental impacts arising from the construction and operation of the Project and associated works that will take place concurrently. The information obtained in the EIA study will contribute to decisions by Director of Environmental Protection (DEP) on:

(i) the overall acceptability of any adverse environmental consequences that are likely to arise as a result of the Project;
(ii) the conditions and requirements for the detailed design, construction and operation of the Project to mitigate against adverse environmental consequences wherever practicable; and
(iii) the acceptability of residual impacts after the proposed mitigation measures are implemented.

1.4 Purpose of this Executive Summary

1.4.1.1 This Executive Summary (ES) summarises the key findings, recommendations, and conclusions of the EIA report for this Project. The ES contains the following information:

- Section 2 presents the purpose and nature of the Project, the benefits of the Project, consideration of alternative options and construction methods for the Project;
- Section 3 presents the key findings of the environmental impact assessment;
- Section 4 describes the proposed environmental monitoring and audit programme for the Project; and
- Section 5 presents the conclusion.

2. PROJECT DESCRIPTION

2.1 Objective and Scope of the Project

2.1.1.1 The Project's objective is to deliver a new road connecting Yuen Long Highway/ Tsing Long Highway and Fanling Highway, linking key new development nodes such as Ngau Tam Mei New Development Area (NTM NDA) and San Tin Technopole (STT). The location of the Project is illustrated in **Figure 1.1**.

2.2 Designated Project under the EIAO

2.2.1.1 The Project comprises various Designated Projects (DPs) in Part I, Schedule 2 of the EIAO as described in **Table 2.1** and illustrated in **Figure 1.2**.

Table 2.1 Schedule 2 Designated Projects in this Project

Schedule 2 Designated Project		Designated Project Element under this Project
Item A.1	A carriageway for motor vehicles that is an expressway, trunk road, primary distributor road or district distributor road	Construction and operation of trunk road with associated primary distributor road and district distributor road at interchanges
Item A.7	A road tunnel or railway tunnel more than 800 m in length between portals	Construction and operation of tunnels including one with a length of around 2.0km between portals
Item I.1	A drainage channel or river training and diversion works located less than 300 m from the nearest boundary of an existing or planned conservation area	Potential temporary diversion during construction of Interchange I/C-02 foundation in Kam Tin River, site formation and road construction near Ngau Tam Mei and Hadden Hill, which locate less than 300m from the nearest boundary of an existing conservation area
Item K.10	A depot for the storage of, or a manufacturing plant for the manufacture of, explosives (as defined by section 2 of the Dangerous Goods Ordinance (Cap. 295))	Temporary explosive magazine site at Tai Shu Ha (Yuen Long) during construction phase
Item Q.1	All projects involving earthworks, dredging works and other building works partly or wholly in an existing or gazetted proposed country park or special area, a conservation area, an existing or gazetted proposed marine park or marine reserve, a site of cultural heritage, and a site of special scientific interest	Earthworks (e.g. at-grade road work, retaining wall construction, slope works, viaduct foundation construction, etc.), and building works partly or wholly in existing "Conservation Area" zone gazetted under Nam Sang Wai OZP no. S/YL-NSW/11 , Kam Tin North OZP no. S/YL-KTN/11, Ngau Tam Mei OZP no. S/YL-NTM/15 and San Tin Technopole OZP no. S/STT/2

2.3 Need of the Project

2.3.1.1 With the progressive development of STT and NTM NDA, traffic congestion is anticipated at San Tin Highway during peak hours in 2036. The Project will alleviate the San Tin Highway traffic congestion by providing an alternative route.

2.3.1.2 In longer term, the transport demand for east-west connectivity within the Northern Metropolis and cross-boundary connections is projected to increase substantially – driven by the progressive development of the Northern Metropolis and the gradual increase on the residential and employment population. The Project, together with the other three sections of NMH, can provide alternative route from New Territories East and West, and alleviate the traffic congestion on Yuen Long Highway, San Tin Highway and Fanling Highway.

2.4 Benefits of the Project

2.4.1.1 The Project can bring the following benefits:

Environmental Benefits and Environmentally Friendly Design

(a) Avoidance of direct impact to ecologically sensitive areas - The recommended alignment and its works footprint have been thoroughly considered and evaluated from alternative alignment options proposed in previous stage studies (**Section 2.6** refers). This process has resulted in the recommended alignment option, which aims to avoid both the direct and indirect potential impacts on recognised sites of conservation importance and other ecologically sensitive areas, such as the Lam Tsuen Country Park (LTCP), the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam Mei. The locations of piers are also carefully designed to be outside the Ngau Tam Mei natural watercourses.

(b) Wetland Habitat Enhancement through Integration and Expansion – Under the original alignment, an interchange, I/C-03, (with associated site office and ancillary facilities) was proposed to the west of Ngau Tam Mei Water Treatment Works (NTMWTW), where scattered and fragmented wetland habitats are currently found. Compared to the original alignment, the recommended alignment has been refined, with the Interchange I/C-03 relocated to the south of NTM NDA. The viaduct level has also been raised as much as practicable to provide greater clearance allowing natural sunlight to reach the habitats underneath the viaduct structures. The relocation of Interchange I/C-03 to the south of NTM NDA will also provide an opportunity to integrate and expand the wetland habitats to the west of NTMWTW. Furthermore, the mainline will be constructed as an elevated viaduct with a limited number of piers, spanning over the wetland habitats to reduce direct ecological impact to this wetland habitats. Taking advantage of the Project implementation, the area will not only be reinstated, but the existing fragmented wetland habitats will also be integrated and enhanced into a connected wetland habitat. The total wetland area will increase from approximately 3 hectares of fragmented ponds to more than 5 hectares of integrated wetland habitats through the incorporation of adjacent village / orchard land. With improved design and management, the enhanced wetland habitat will support higher biodiversity, greater ecological value and increased carrying capacity. It will also offer a suitable foraging environment for various relevant species, including waterbirds.

(c) Allowing Wildlife Movement by Adopting Intermittent At-grade and Viaduct Design – A section of the alignment runs adjacent to the woodland habitat at the edge of the LTCP near Long Ha. The alignment profile has been raised as far as practicable, taking into account site constraints such as CLP cables, pylons and traffic design requirements. The design has been optimised to avoid encroachment into LTCP while maximising the length of the viaduct section. Each of the at-grade sections is no greater than ~350 m so as to avoid creating potential hurdles to wildlife movement. The space beneath the viaduct sections provides an open corridor to facilitate wildlife movement to and from LTCP. For the at-grade sections, retaining wall on the slope would be constructed to avoid wildlife roadkill.

(d) Viaduct Level Arrangement and Carriageway Separation for Enhancing Sunlight Penetration and Ecological Benefits – The Project includes slip roads crossing the Kam Tin River, where the viaduct level has been adjusted to provide headroom of about 10~15m and minimise shading effects on the water body and associated habitats. In the vicinity of NTMWTW, to accommodate the required clearance from the existing CLP cable, the headroom beneath the viaduct is approximately 10m for the northbound and ranges from 5m to 10m for the southbound. The dual carriageways are separated before entering the tunnel portal. The resulting gap between the carriageways enables natural sunlight to reach the ground below and is not expected to cause adverse ecological impacts. Furthermore, the positioning of piers have been refined to avoid encroachment into natural watercourses. This approach helps to minimize potential water quality and ecological impact.

(e) Adoption of Non-dredged Method for Construction Works at Kam Tin River - The significance of ecological value of Kam Tin River is well recognised. Given the large span of the viaduct crossing Kam Tin River, the pile /pier support within the river is inevitable. The construction of viaduct pier/ pile will adopt the non-dredging method. For example, if bored pile construction is adopted, steel casing could be adopted to isolate the surrounding riverbed. The material within the steel casing will be excavated within the casing for placing reinforcement and concrete of the pile. This minimises the disturbance to the river.

(f) Synergistic Woodland Enhancement near She Leng – A Woodland compensation of approximately 17 ha is proposed near She Leng, which is currently characterised by grassland/shrubland of low or low to moderate ecological value. The site is also adjacent to proposed woodland compensation areas under other projects, including Lok Ma Chau Loop, STT and Development of the Loop – Eastern Connection Road (ECR). Woodland compensation at this location will integrate the overall compensatory woodland area and improve habitat quality and ecological connectivity with adjacent woodland habitats.

(g) Minimisation of slope cutting through adoption of elevated alignment – Given the hilly terrain to the south of Ngau Tam Mei New Development Area (NTM NDA), the road alignment has been elevated. This adjustment not only minimises the scale of site formation works, but also takes into account other constraints, such as the clearance requirement for CLP power cables. Additionally, a larger headroom would provide greater clearance between the

structure and the habitats so as to allow natural sunlight to shine on the habitats underneath the viaduct structures.

(h) Minimisation of slope cutting through soft ground tunneling at portals – When the alignment passes through Ngau Tam Shan and Hadden Hill, soft ground tunnelling has been adopted at portals. Specifically, before the tunnel excavation reaches bedrock, soft soil stratum will be improved with injection grouting around the tunnel profile. Temporary support such as pipe roofing and steel lattice girder/rib will be installed during the tunnel excavation by mechanical method. With the adoption of soft ground tunnelling, the extent of site formation is significantly reduced, which helps retain a larger area of natural terrain and minimise the generation of construction and demolition (C&D) materials.

(i) Minimisation of slope cutting through construction of retaining walls and soil nailing – For sections where extensive cut slopes would otherwise be required, retaining walls / soil nails or combination of both may be proposed. This measure aims to reduce the extent of slope excavation, allow for steeper slope gradients, and thereby minimise the project's environmental footprint, as well as acting as a barrier to prevent wildlife roadkill.

(j) Avoidance of direct impacts on Sites of Archaeological Interest (SAI) – Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI to avoid potential direct impacts on the SAI.

(k) Adopt Drill and Blast (D&B) method for tunnel construction – Adopt D&B method over Tunnel Boring Machine (TBM) method for tunnel construction to minimize the generation of C&D materials as much as practicable.

(l) Diverting the existing traffic flow between northeast and northwest of the New Territories, lowering vehicular emissions from idling or congested traffic and shortening travel times and results in further reducing fuel consumption, alleviating potential air quality and noise impact during operation phase.

Social Benefits

(m) Serving the traffic demand for better east-west connectivity within the Northern Metropolis and cross-boundary connections.

(n) Providing more connectivity to the existing road and the development nodes in the Northern Metropolis, which improves transport efficiency and enhances connectivity.

(o) Encouraging and enabling the public to easily access various land-based boundary control points (BCPs) via major roads and railways, boosting the connectivity and synergy effect among these BCPs as well as the benefits of the cross-boundary infrastructure.

(p) Supporting 'infrastructure-led' growth at the development nodes and creating job opportunities.

2.4.1.2 Due consideration has been given above to address the environmental challenges encountered in the design of the Project. Environmental protection will be prioritised and fulfilled to the greatest extent possible. Various key actions have been implemented to address the environmental challenges and summarised in **Table 2.2**.

Table 2.2 Summary of Alternative Mitigation Measures

Alternative Mitigation Measures	Details	Environmental Benefits	Environmental / Engineering Disbenefits
Avoidance of Recognised Sites of Conservation Importance and Other Ecologically Sensitive Areas	• Avoid both the direct and indirect potential impacts on recognised sites of conservation importance and other ecologically sensitive areas, such as the LTCP, the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam Mei • Relocate I/C 03 to the south of NTM NDA	• Avoid direct ecological impact on the recognised sites of conservation importance and other ecologically sensitive areas	• More interface with existing CLP, and Express Rail Link (XRL) at the relocated I/C-03
Avoidance of Ngau Tam Mei SAI	• Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI	• Avoid archaeological impact on Ngau Tam Mei SAI	• Nil
Minimisation of C&D materials	• Minimisation of slope cutting through adoption of elevated alignment to the south of NTM NDA • Minimisation of slope cutting through soft ground tunnelling at portals of Ngau Tam Shan and Hadden Hill • Minimisation of slope cutting through construction of retaining wall and soil nailing to allow for steeper slope gradient and reduce extent of excavation • Adoption of D&B method for tunnel construction over TBM method	• Minimise the extent of slope cutting required and thus minimising the generation of C&D materials • Minimise C&D materials generated during tunnel construction	• Nil

Alternative Mitigation Measures	Details	Environmental Benefits	Environmental / Engineering Disbenefits
Minimisation of water quality and ecological impact	• Adopt non-dredged method for construction works at Kam Tin River	• Minimise the disturbance to the water flow regime of Kam Tin River and ecological impact	• Less flexibility on construction methods, leading to associated cost and programme implication
Minimisation of ecological impact	• Adopt viaduct with limited number of piers at the wetland habitat to the west of NTMWTW • Adopt intermittent at-grade and viaduct design for the alignment near LTCP • Adjust the viaduct level of the alignment to provide sufficient headroom	• Minimise habitat loss due to piers • Minimise potential ecological impact due to shading effect of the viaduct • Facilitate wildlife movement to and from LTCP	• By maximising the elevated viaduct section and headroom, clearance with overhead CLP cable is reduced and leading to construction constraint

2.5 Consideration of "With" and "Without" Project Scenarios

2.5.1 Scenario "with" Project

2.5.1.1 Upon the implementation, the Project will provide an alternative route from Yuen Long Highway / Tsing Long Highway to Fanling Highway which will alleviate the traffic pressure on existing road network with progressively increasing traffic demand.

2.5.1.2 It will also link up the development nodes with provision of direct access to areas such as NTM NDA, STT and Kwu Tung North New Development Area (KTN NDA) to significantly improve transport efficiency of the strategic road network. The enhanced accessibility will in turn drive and expedite the implementation of development projects within these areas and fully unleashes the development potential.

2.5.1.3 The public and visitors could make use of the developed major road network to conveniently travel to the various land-based boundary control points in the Northern Metropolis. It will effectively improve the cross-boundary road connection for passengers and goods in Hong Kong.

2.5.1.4 The Project would constitute one of the four sections of NMH which effectively connects the New Territories East and West. The driving distances within the Northern Metropolis will be greatly reduced after the commissioning of the NMH. The reduced traffic time would save the energy consumption and the improved traffic efficiency would minimise potential air quality impact and noise impact to the sensitive receivers.

2.5.1.5 In addition, the Project incorporates ecological-friendly design measures, which help preventing wildlife from entering carriageways and reduce roadkill risk. The Project also enables enhanced ecological connectivity through planned habitat

creation, such as wetland compensation areas that complement existing ecological resources at the area to the west of the NTMWTW, thereby improving long-term habitat continuity and supporting biodiversity.

2.5.2 Scenario “without” Project

2.5.2.1 Without the implementation of the Project, traffic congestion would be expected at road section of San Tin Highway in year 2036 during peak hour with the progressive development of development nodes such as STT and the NTM NDA.

2.5.2.2 The connectivity of development nodes will not be as direct and convenient as with the Project which in turn limits the potential of development area.

2.5.2.3 The overstress of existing road network would cause traffic congestion which in turn induces potential air quality and noise impacts to the sensitive receivers.

2.5.2.4 Without the Project, the east-west traffic connection in Northern Metropolis would heavily rely on the existing San Tin Highway which will operate beyond its capacity.

2.6 Consideration of Alignment Options

2.6.1 General

2.6.1.1 The alignment under the Project consists of 5 interchanges (I/C-02 to I/C-06) from Yuen Long Highway / Tsing Long Highway to Fanling Highway. Interim interchanges (I/C-03 to I/C-05) are provided to connect some development nodes and existing roads. The I/C-02 and I/C-06 will reserve stub ends for connection to the future Tin Shui Wai section and Kwu Tung section of NMH respectively.

(a) I/C-02 will reserve stub end at west for future connection to Tin Shui Wai Section of NMH.

(b) I/C-02 to I/C-03 connects Yuen Long Highway/ Tsing Long Highway and NTM NDA

(c) I/C-03 to I/C-04 connects NTM NDA and STT

(d) I/C-04 to I/C-05 connects STT and Fanling Highway

(e) I/C-05 to I/C-06 connects Fanling Highway and KTN NDA

(f) I/C-06 will reserve stub end at north for future connection to Kwu Tung Section of NMH.

2.6.1.2 Original Alignment Option was initially derived from the SSMR FS under Agreement No. CE7/2018(TT).

2.6.1.3 After commencement of the Investigation stage, more detailed analysis has been conducted, and several alignment options have been further explored aiming to optimise the scheme and to enhance environmental benefits.

2.6.1.4 With preliminary review and screening, three alignment options (inclusive of the Original Alignment Option) have been shortlisted and taken forward for evaluation from perspectives of site constraints, traffic performance, engineering

constructability, social acceptance, environmentally friendliness, etc to suggest the recommended alignment.

2.6.1.5 Alignment Option 3 is recommended after robust considerations. **Figure 1.3** (**Figure 1.3.1-1.3.4**) delineates the three alignment options. **Table 2.3** below presents the considerations of alignment options with elaboration.

Table 2.3 Alignment Options Comparison

Alignment Option 1 (Original Alignment)	Alignment Option 2	Alignment Option 3 (Recommended Alignment)
Pros ✓ • Mostly tunnel and minimise interface with aboveground facility (e.g.CLP Pylon). • Minimise ecological impact by tunnel form near I/C-02.	Pros ✓ • Mostly tunnel and minimise interface with aboveground facility (e.g.CLP Pylon). • Minimum alignment length • Minimise ecological impact by tunnel form near I/C-02.	Pros ✓ • Better traffic connectivity with direct connection to Route 3 via Tsing Long Highway. • Mostly aboveground and minimise interface with underground facility (e.g. Guangzhou-Shenzhen-Hong Kong Express Rail Link (XRL), existing water tunnel). • Lane changing and weaving flexibilities are allowed. • Avoid direct ecological impact on the wetland habitat near Nam Sang Wai. • Completely avoids the LTCP. Minimise ecological impact. • Lesser the C&D materials when compare with Option 1 & 2, due to reduction in tunnel length. • Lower capital and re-current cost with shorter tunnels. • Direct connection to the planned residential sites of NTM NDA at I/C-03. • Signalised junctions at I/C-04 and I/C-06 minimise interchange footprint while in turn reduce cost, land take and environmental impact.
Cons ✕	Cons ✕	Cons ✕

Alignment Option 1 (Original Alignment)	Alignment Option 2	Alignment Option 3 (Recommended Alignment)
• I/C-02 depressed slip road and tunnel work construction have significant impact on San Tin Highway, Castle Peak Road, and San Tam Road. • TBM Method near I/C-02 before entering LTCP. High risk of tunnelling in shallow ground cover which may induce ground settlement and affect the population nearby and heavy traffic on San Tin Highway. • Temporary deep and large area excavation in wetland habitat at Nam Sang Wai for deep launching/retrieval shaft for TBM launching which involve dredging and dewatering. Launching the TBM from the wetland at Nam Sang Wai would cause direct ecological impact to the wetland area. • Large extent of tunnel parallel run and close to water tunnel which may have cost and programme implications. • Lane change and weaving opportunities are limited due to tunnel portions. • No direct connection to Tsing Long Highway. The connection to Tsing Long Highway will have to via existing San Tin Highway. • Generate more C&D materials due to long tunnel. • High capital and re-current cost with more tunnels. • Larger interchange footprint at I/C-04 and I/C-06.	• I/C-02 depressed slip road and tunnel work construction have significant impact on San Tin Highway, Castle Peak Road, and San Tam Road. • TBM Method near I/C-02 before entering LTCP. High risk of tunnelling in shallow ground cover which may induce ground settlement and affect the population nearby and heavy traffic on San Tin Highway. • Temporary deep and large area excavation in wetland habitat at Nam Sang Wai for deep launching/retrieval shaft for TBM launching which involve dredging and dewatering. Launching the TBM from the wetland at Nam Sang Wai would cause direct ecological impact to the wetland area. • Large extent of tunnel parallel run and close to water tunnel which may have cost and programme implications. • Lane change and weaving opportunities are limited due to tunnel portions. • No direct connection to Tsing Long Highway. The connection to Tsing Long Highway will have to via existing San Tin Highway. • Generate more C&D materials due to long tunnel. • High capital and re-current cost with more tunnels. • Affects the land use planning of the NTM NDA and the STT.	• Slightly longer alignment length and thus a very slight increase in driving time. • Potential impact on flight path near I/C-02.

Alignment Option 1 (Original Alignment)	Alignment Option 2	Alignment Option 3 (Recommended Alignment)
	• Larger interchange footprint at I/C-04 and I/C-06.	

2.7 Construction Methodologies for Key Design Elements

2.7.1.1 The key construction elements of the Project include construction of at-grade roads, viaduct, tunnel, administration, portal plant buildings / plant room and satellite control buildings, retaining wall / slope strengthening.

2.7.1.2 The selection of construction methodologies has been carefully considered to minimise environmental impacts, optimise constructability, and reduce construction duration.

2.7.1.3 Possible construction methods which have been investigated for the construction of the Project and the reasons for selection are summarised in **Table 2.4** below.

Table 2.4 Summary of Possible Construction Methods for the Project

Construction element	Possible construction method	Reason
At-grade roads	Cut and fill method	Due to the hilly terrain to the south of Ngau Tam Mei, the cut and fill method is proposed to cope with the level difference. The cut material from higher terrain will be used to fill lower-lying and achieve cut-and-fill balance.
Viaduct	Precast prestressed concrete construction method	Better control of quality and workmanship for works in fabrication yard. This also reduces the generation of C&D materials on site and minimises construction air quality and noise impact
Tunnel	Drill-and-blast	Adopting D&B method can avoid excessive over-excavation and C&D waste. Small section with Mechanical Break will be adopted where poor ground conditions or sensitive receivers are encountered. TBM is considered not preferrable considering the cost, constructability, programme, environmental impact.
Construction works at Kam Tin River	Provision of temporary steel decking and non-dredged piling construction method with staged construction	To minimize the ecological impact and water quality impact at Kam Tin River
Construction of Administration, Portal Plant Building / Plant Room and Satellite Control Buildings	Conventional reinforced concrete construction method	The buildings are small in scale and low rise. The construction method suits for site in low accessibility. Precast concrete elements would be explored wherever possible to minimise on-site work and the associated environmental impacts, including the generation of C&D

Construction element	Possible construction method	Reason
		materials, construction air quality and noise impact, and other disturbance to surrounding environment.
Construction of retaining wall / slope strengthening	Conventional reinforced concrete retaining wall construction, tie back and slope strengthening by soil nail	The construction method suits the stage by stage slope cutting and site formation works, and minimise the extent of slope formation.

2.8 Implementation Programme

2.8.1.1 The Project is tentatively scheduled to be ready for tender invitation in 2027. Subject to the further planning and actual implementation of the Project, advance work and related study might commence in the same year or in 2028.

2.8.1.2 For the purposes of Environmental Impact Assessment study, construction of the Project is anticipated to commence tentatively in 2027 with a target to commission in or before 2036. The below phase division is based on the grouping of major construction activities for environmental impact assessment purposes. In practice, due to the complexity of the Project, certain works across different phases may overlap or be carried out consecutively across phases. The actual construction sequence and schedule will be further optimised based on detailed engineering design.

Table 2.5 Tentative Construction Programme

Phase	Tentative Construction Programme	Key Activity Description
Phase 1	Q3 2027 to Q4 2030	Construction advance works and initial civil engineering, including site clearance, at-grade road site formation (comprising temporary haul roads and general site formation), and portal site formation. Later in this phase, main construction works will include like at-grade road construction (including road surfacing), viaduct construction (involving pier foundation excavation), and tunnel excavation etc.
Phase 2	Q1 2031 to Q4 2033	Full scale execution of main civil works which covers the construction of at-grade roads, viaduct construction, tunnel excavation, construction of ancillary structures, etc (e.g., administration buildings, portal plant buildings/ plant room etc.)
Phase 3	Q1 2034 to Q4 2036	Completion of all remaining construction works, including but not limited to finishing of at-grade roads, viaducts, tunnels, and ancillary buildings. It also encompasses the installation, commissioning and testing of associated electrical and mechanical

Phase	Tentative Construction Programme	Key Activity Description
		(E&M) systems, as well as site reinstatement works, etc.

3. KEY FINDINGS OF THE ENVIRONMENTAL IMPACT ASSESSMENT

3.1 Approach to the EIA

3.1.1.1 The EIA process provides a means of identifying, assessing and reporting the environmental impacts associated with the construction and operation of the Project based on the engineering design information available at this stage. Throughout the EIA process the potential environmental effects of various design options and develop alternatives as well as mitigation measures to be incorporated into the design, construction and operation of the Project are identified. Mitigation measures have been recommended to avoid some potential environmental impacts, or to minimise/mitigate impacts to acceptable levels.

3.1.1.2 The findings of this EIA Study have determined the likely nature and extent of the following environmental impacts predicted to arise from the construction and operation of the Project:

- Air Quality Impact
- Noise Impact
- Water Quality
- Waste Management Implications
- Land Contamination
- Ecological Impact
- Fisheries Impact
- Landscape and Visual Impact
- Cultural Heritage Impact
- Hazard to Life

3.2 Air Quality Impact

3.2.1.1 Potential air quality impacts associated with the construction and operational phases of the Project have been assessed in accordance with the requirements given in Clause 3.5.3 and Appendix B of the EIA Study Brief, as well as the criteria and guidelines as stated in Annexes 4 and 12 of EIAO-TM. The assessment area for air quality impact assessment is defined by a distance of 500 m from the boundary of the Project and its associated works as identified in the EIA study.

3.2.1.2 Potential air quality impact from the construction works of the Project would mainly be related to construction dust from site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials, wind erosion of exposed area, as well as use of on-site powered mechanical equipment, and emissions from construction vehicles .

3.2.1.3 Construction activities of the concurrent projects within 500 m assessment area would also contribute to cumulative air quality impact. With the implementation of mitigation measures specified in the *Air Pollution Control (Construction Dust) Regulation*, together with the recommended control measures including frequent watering on active works areas, exposed areas and unpaved haul roads and other site management measures such as good site practices, use of electric vehicles and Environmental Monitoring and Audit (EM&A) programme, no adverse air quality impact on air sensitive receivers (ASRs) would be anticipated during the construction stage.

3.2.1.4 Cumulative air quality impacts during the operational phase of the Project were assessed by considering the project-induced emissions, all relevant emission sources within the 500 m assessment area, as well as background contributions. The quantitative modelling prediction results concluded that the cumulative NO_2, RSP, and FSP concentrations at all existing and planned ASRs would comply with the prevailing Air Quality Objectives (AQOs). Thus, no adverse air quality impact during the operation phase of the Project on the ASRs would be anticipated.

3.3 Noise Impact

3.3.1.1 Potential noise impacts associated with the construction and operation phases of the Project have been assessed in accordance with the requirements given in Clause 3.5.4 and Appendix C of the EIA Study Brief, as well as the criteria and guidelines as stated in Annexes 5 and 13 of the EIAO-TM. The assessment area for construction noise, fixed noise, and road traffic noise impact assessments was defined by a distance of 300 m from the boundary of the Project and its associated works as identified in the EIA study.

3.3.1.2 Qualitative construction noise assessment of the Project has been conducted. With the adoption of proposed noise mitigation measures, including use of Quality Powered Mechanical Equipment (QPME), use of quieter equipment / method, use of higher surface density site-hoarding, use of noise barrier / enclosure, etc., adverse noise impact arising from construction works of the Project is not anticipated.

3.3.1.3 Construction Noise Management Plan (CNMP), which contains a quantitative construction noise impact assessment, mitigation measures and monitoring and audit programme, will be submitted to EPD before the tender invitation and commencement of construction works, with details on the construction method, plant inventory, recommended noise mitigation and implementation details of the mitigation measures in order to minimise the construction noise impact to comply with the EIAO-TM criteria.

3.3.1.4 A quantitative assessment of road traffic noise impacts has been conducted in accordance with the requirements outlined in the EIA Study Brief. Under the unmitigated scenario, the overall predicted noise levels at some of the representative noise-sensitive receivers (NSRs) would exceed the noise criteria. To mitigate potential road traffic noise impacts, direct at-source measures such as Low Noise Road Surfacing (LNRS) and noise barrier have been considered, while at-receiver mitigation measures, including acoustic windows and acoustic balconies, are to be implemented by the planned NSRs. With the proposed noise mitigation measures in place, the Project roads noise levels at all these NSRs would comply with the relevant noise criteria or the Project roads contributions to the overall noise levels would be insignificant, i.e. less than 1.0 dB(A). As such, no adverse road traffic noise impact arising from the Project is anticipated and no further mitigation measures would be required.

3.3.1.5 A qualitative assessment on impacts from fixed noise sources has been conducted. The impact from the planned fixed noise sources could be effectively mitigated by implementing at-source noise control measure during the detailed design stage, no adverse cumulative impact from the existing fixed noise sources, the proposed fixed noise sources under the Project. Fixed Noise Sources Management Plan (FNMP) would be submitted separately to EPD for agreement prior to issuance of tender and commencement of construction of the proposed fixed noise source to recommend detailed fixed noise mitigation measures and ensure noise compliance. The FNMP would also include the prevailing background noise survey, the quantitative fixed noise source impact assessment, and monitoring and audit programme.

3.4 Water Quality Impact

3.4.1.1 The water quality impact assessment (WQIA) was conducted in accordance with the requirements given in Clause 3.5.5 and Appendix D of the EIA Study Brief as well as the criteria and guidelines as stated in Annexes 6 and 14 of the EIAO-TM. The assessment area for this water quality impact assessment includes areas within 500 m from the boundary of the Project and its associated works as identified in the EIA study and covers Deep Bay Water Control Zone (WCZ) as designated under the Water Pollution Control Ordinance (WPCO).

3.4.1.2 The potential water quality impacts associated with the construction works include construction works in Kam Tin River, wastewater generated from general construction activities, construction site runoff, construction works near watercourses, removal and/or diversion of watercourses and/or ponds, accidental spillage of chemicals, sewage from construction workforce, groundwater infiltration arising from tunnel boring, groundwater from potential contaminated areas, contaminated site runoff and wastewater from land decontamination. Given these impacts could be mitigated and controlled by implementing the recommended mitigation measures, no adverse water quality impact during construction phase is expected. Regular site inspections and water quality monitoring should be undertaken routinely to inspect the construction activities and works area to ensure the recommended mitigation measures are properly implemented.

3.4.1.3 During operation phase, potential water quality impacts are related to surface runoff generated from the new paved areas as well as the change in flow regime and water quality impacts due to the presence of bridge piers in Kam Tin River. Provided that the recommended mitigation measures for the drainage system are properly implemented, the associated water quality impacts are expected to be minimal and acceptable. Considering that the proposed bridge piers will be oriented with the existing piers of the Tsing Long Highway / Yuen Long Highway, no significant change in flow regime and water quality in Kam Tin River is expected and therefore, no adverse water flow and water quality impacts on Kam Tin River is expected.

3.4.1.4 All sewage generated from the operation staff would be collected and discharged to the public sewerage system and diverted to sewage treatment works for further treatments. No adverse water quality impact is therefore expected.

3.5 Waste Management Implications

3.5.1.1 The waste management implication assessment has been conducted in accordance with the criteria and guidelines as stated in the requirements given in Section 3.5.6 and Appendix E of the EIA SB, as well as Annexes 7 and 15 of the EIAO-TM.

3.5.1.2 During construction phase, wastes generated from the Project include C&D materials and excavated sediments from construction activities, chemical wastes from maintenance and servicing of construction plants and vehicles, as well as general refuse from workforce. It is estimated that approximately 5,495,370 m^3 of inert materials and 174,220 m^3 of non-inert materials will be generated from site clearance, site formation, tunnel excavation, pier foundation excavation, road surfacing, underground utilities and wetland compensation excavation works. Of the inert C&D materials, around 109,910 m^3 is expected to be reused on-site as backfilling materials and around 5,385,470 m^3 is expected to be delivered to Public Fill Reception Facility, subject to the allocation by the Public Fill Committee. The non-inert C&D materials would be reused and recycled before disposal to landfill. It is expected that approximately 11,080 m^3 of excavated sediment, around 17,550 tonnes of general refuse during the whole construction period and chemical waste in the order of few hundred litres per month would be generated during the construction phase. With the implementation of recommended good site practices and mitigation measures, no adverse environmental impacts (including potential hazard, air and odour emissions, noise and wastewater discharges) would be anticipated during the construction phase.

3.5.1.3 The main types of wastes generated from the operation of the Project are general refuse and chemical wastes. With the implementation of management control practices, no adverse environmental impact would be anticipated.

3.6 Land Contamination

3.6.1.1 Assessment on potential of land contamination has been conducted in accordance with the guidelines stated in Annex 19 of the EIAO-TM, as well as the requirements given in Section 3.5.7 and Appendix F of the EIA Study Brief.

3.6.1.2 Site appraisals, in the form of desktop review and site walkovers, had been carried out between September 2025 and December 2025 to identify the past and current potentially contaminating land uses within the Project Site. Based on the site appraisal, 30 potentially contaminated sites, 15 sites suspected to be used for industrial purposes and 23 sites with low land contamination potential were identified within the Project Site.

3.6.1.3 Further land contamination assessments, including further site appraisal and submission of Contamination Assessment Plan(s) (CAP(s)), should be conducted for the whole Project Site at a later stage of the Project when site access is available to confirm the existing land uses / activities, identify the presence of any potential contamination sources, and address any new contamination issues. The associated Site Investigation (SI) works and any necessary remediation action are recommended to be carried out after operation of concerned site(s) has ceased but prior to the commencement of construction works. The recommended further assessment and remediation works, including the submission of CAP(s), Contamination Assessment Report(s) (CAR(s))/Remediation Action Plan(s) (RAP(s)) and Remediation Report(s) (RR(s)) would follow the relevant Guidance Manual, Guidance Note and Practice Guide.

3.6.1.4 With reference to the *Preliminary Feasibility Study on Developing the New Territories North*, the *Geochemical Atlas of Hong Kong* published by the Geotechnical Engineering Office of Civil Engineering and Development Department in 1999 and the Hong Kong Environmental Database (HKED) by EPD, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared, in consultation with EPD, and submitted to EPD for endorsement.

3.6.1.5 With the implementation of the recommended further works for the Project, any soil/groundwater contamination would be identified and properly remediated prior to the construction works. No insurmountable land contamination impacts to the Project are therefore anticipated.

3.7 Ecology

3.7.1.1 Ecological impact assessment for the Project has been conducted in accordance with the requirements given in Clause 3.5.8 and Appendix G of the EIA Study Brief, the criteria and guidelines of the EIAO-TM Annexes 8 and 16 as well as relevant EIAO Guidance Notes.

3.7.1.2 The Assessment Area includes areas within a 500 m from the boundary of the Project and its associated works as identified in the EIA study. Ecological field surveys have been conducted over a 12-month period covering both wet and dry seasons. Several sites of conservation importance were identified within the Assessment Area, including Mai Po Inner Deep Bay Ramsar Site, Ramsar Site Priority Site for Enhanced Conservation, LTCP, Conservation Area (CA), Wetland Conservation Area (WCA), Wetland Buffer Area (WBA), and the proposed Nam Sang Wai Wetland Conservation Park (NSW WCP). Other ecologically sensitive receivers included Inner Deep Bay and Shenzhen River Catchment Important Bird Area (IBA), night roosts at Sha Po and Ko Po Road, planned bat shelter at Pok Wai, Sha Po Marsh, Yuen Long Bypass Floodway Engineered Wetland, planned compensation wetland for Ngau Tam Mei Development Area, West Rail Compensatory Wetland, Kam Tin Main Drainage Channel, and Planned Wetland Compensation Site at Kam Tin. The Project Boundary encroaches into small areas of WCA, WBA, and Deep Bay Wetland outside Ramsar Site Priority Sites for Enhanced Conservation, but encroached habitats are limited to man-made habitats including developed area/wasteland, roadside plantation, and small section of modified watercourse, which are all subject to regular disturbance (e.g. traffic and human activities along existing highway). For CA, impact may arise from the loss of woodland habitats, which will be compensated with a Woodland Compensation Plan.

3.7.1.3 A total of 15 habitat types were identified during the ecological survey within the 500 m Assessment Area, which included woodland, mixed woodland, shrubland, grassland, plantation, village/orchard, developed area/wasteland, agricultural land, seasonally wet grassland, marsh/reed, pond, natural watercourse, semi-natural watercourse, modified watercourse, mitigation wetland. These habitats are generally considered to be of “low” or “low to moderate” ecological values, except for woodland, ponds within WCA and WBA, and Kam Tin Main Drainage Channel which are considered as “moderate” ecological value, as well as Sha Po Marsh (outside of Project Boundary) which is considered as “moderate to high” value.

3.7.1.4 Direct impacts arising from the Project include direct loss of habitats and vegetation due to the proposed permanent structures and the associated construction activities. Avoidance and minimisation measures (e.g. refinement of alignment and establishment of no-work-zone during construction activities, piling works at the Kam Tin Main Drainage Channel (KTMDC) would be scheduled to conduct during high tidal level (i.e., above 1.0 mCD) during dry season when the riverbed of KTMDC would be mostly submerged) have been considered to reduce and minimise the direct permanent and temporary impacts to these habitats. Additionally, to further minimise potential impacts to the ecologically sensitive habitats (i.e., KTMDC and ponds near NTMWTW), phasing of construction works would be adopted in these areas under the Project. This phased construction in KTMDC and ponds near NTMWTW aims to reduce the scale of disturbance and the total area of construction activities. The area occupied by actual construction (i.e. pond habitat near NTMWTW and KTMDC)

would not exceed half of the designated temporary works area at any given time. The remaining half would be maintained as pond habitat or watercourse. The impacted areas from temporary works will be reinstated to their pre-existing condition as much as practicable before commencing construction on the remaining half. Woodland and wetland compensation measures have also been recommended to mitigate for the inevitable loss of ecologically sensitive habitats. Direct impact may also potentially arise from the potential direct injury / mortality on species of conservation importance within the Project Boundary, in particular species with low mobility including flora species Cycad-fern and Incense Tree, as well as firefly species Motschulsky's Starworm. Pre-construction site check with associated transplantation and translocation exercises were recommended.

3.7.1.5 On the other hand, indirect impacts such as disturbance impact to adjacent recognised sites of conservation importance, ecologically sensitive resources (such as Sha Po Night Roost and a planned bat shelter), as well as night-time disturbance and potential water quality and water flow impact may be induced by the construction and operation of the Project. Measures such as provision of screening, use of directional lighting, general good site practice and other noise, air and water quality mitigation measures are recommended. Moreover, specific measure for the Sha Po Ardeid Night Roost is also recommended, including a 100m Buffer Area for the night roost, where noisy works would be avoided from one hour before sunset to one hour after sunrise. Use of non-transparent or non-glaring materials are also recommended to avoid and minimise the potential direct impacts to the habitats and the associated wildlife within and adjacent to the Project Site.

3.7.1.6 With the implementation of the aforementioned mitigation measures, no unacceptable ecological impacts are anticipated to arise from the Project on the ecological resources recorded within the Project Boundary and the Assessment Area.

3.8 Fisheries

3.8.1.1 Potential impacts on fisheries have been assessed in accordance with the requirements given in Clause 3.5.9 and Appendix H of the EIA Study Brief as well as the criteria and guidelines as stated in Annexes 9 and 17 of the EIAO-TM.

3.8.1.2 The fisheries impact assessment has been conducted based on the information gathered from literature review and the site checks. The results showed that the assessment area contains a mixture of active, inactive, and abandoned fishponds, with active aquaculture mainly concentrated in NTM area and to the west of San Tin Highway. Direct permanent loss and temporary loss of fishponds were identified in NTM area while a few were located in Ki Lun Tsuen area. In terms of permanent loss, active fishponds (1.5 ha), inactive fishponds (0.2 ha) and abandoned fishponds (0.7 ha) would be impacted during the construction phase. Considering the fishpond area involved is small in terms of the overall fishpond areas in Hong Kong (i.e. less than 1% in total), the fisheries impact arising from the Project is therefore anticipated to be low.

3.8.1.3 Potential indirect impacts, such as deterioration of water quality, bund instability, or access blockage during construction, may temporarily affect pond operations. However, with the implementation of standard site practices and mitigation measures including bund protection, runoff control, and provision of alternative access, no unacceptable impact is anticipated. During operation, the design of the road alignment ensures that access to fishponds will be maintained, while the

surface run-off will be properly managed through the road drainage system. Overall, the Project is not expected to cause significant impacts to fisheries resources or activities and hence no fisheries-specific monitoring is necessary.

3.9 Landscape and Visual Impact

3.9.1.1 Potential landscape and visual impacts have been assessed in accordance with the requirements given in Clause 3.5.10 and Appendix I of the EIA Study Brief as well as the criteria and guidelines as stated in Annexes 10 and 18 of the EIAO-TM.

3.9.1.2 The Project will inevitably result in some landscape and visual impacts during construction and operational phases. These impacts were minimised through the consideration on minimization of works areas, incorporation of aesthetic external designs and landscape treatments in the proposed infrastructures.

3.9.1.3 Tree compensation within the site in a form of woodland mix tree planting on slopes or heavy standard tree planting in amenity area would be explored. Tree compensation requirement of 1:1 in terms of number should be carried out as far as practicable. Exact number of trees to be retained, transplanted and removed to be determined under the Tree Preservation and Removal Proposals (TPRPs) in accordance with DEVB TC(W) – No. 4/2020 – Tree Preservation. Among the approx. 67,500 nos. of existing trees within the 100m assessment area, it is estimated that there are approx. 9,050 nos. of trees (excluding *Leucaena leucocephala* of which TPRP is not required for removal of undesirable species) would be unavoidably affected. The exact quantity, tree condition, proposed recommendations as well as receptor locations of trees to be transplanted shall be further reviewed in the formal TPRP submission, which to be prepared and submitted in the Detailed Design stage.

3.9.1.4 Under this Project, the existing landscape resources, including vegetation, marsh or wetland and fishpond would be affected in multiple scale subject to the proposed works. Vegetation found within the Landscape Resources (LRs) and Landscape Character Areas (LCAs) would either be temporarily or permanently loss. After the completion of works, landscape mitigation measures such as tree compensation where space is available, reinstatement of affected landscaping area or fishpond in like for like basis and provision of buffer screen planting would be implemented. With the implementation of these mitigation measures, it is considered that residual impacts of the LRs and LCAs would be reduced to **Negligible** to **Slight** in the operational phase.

3.9.1.5 In terms of the visual impact, considered the connecting alignment to the existing infrastructures and structural mass of proposed road works, the visual impact to the identified key public viewers would be ranging from Negligible to Moderate.

3.9.1.6 Considering the scale and nature of the Project, it would inevitably result in certain levels of residual landscape impacts and visual impacts in relation to the loss of hillside vegetation from hilltop and from ground level. Nevertheless, the residual landscape impacts are localized and limited to the project extent only without affecting existing community, while the visual impacts are confined within the visual envelope either involving few to medium numbers of public viewers along hiking trail and paths. With the implementation of the proposed landscape and visual mitigation measures, the overall landscape residual impacts would be from Negligible to Slight in operational phase, and the overall visual impacts would be from Negligible to Moderate in operational phase. With full implementation of the

recommended mitigation measures, unacceptable residual landscape impacts and visual impacts are not expected.

3.10 Cultural Heritage

3.10.1.1 The Cultural Heritage Impact Assessment (CHIA), which covers Built Heritage Impact Assessment (BHIA) and Archaeological Impact Assessment (AIA), has been conducted in accordance with the requirements given in Clause 3.5.11 and Appendix J of the EIA Study Brief.

Built Heritage

3.10.1.2 No declared monument, proposed monument, new item for grading assessment and Government historic site identified by AMO is located within the assessment area. No direct impact on built heritage would therefore be anticipated during both the construction and operational phases.

3.10.1.3 Five (5) grade 2 historic buildings and eight (8) grade 3 historic buildings are located within the assessment area of 300m from the Project Boundary. Given the considerable distance between the identified graded historic buildings and the Project Boundary, no adverse impact on these graded historic buildings would be anticipated during the construction and operational phases.

Archaeology

3.10.1.4 One (1) Site of Archaeological Interest (SAI), namely Ngau Tam Mei SAI (SAI38) is located within the assessment area and it is partly located within the Project Boundary. According to the latest construction works arrangement, the works that will be carried out within the SAI would only be those temporary works (such as filling up the area for haul road use) that are non-intrusive in nature and would cause no sub-soil disturbance. Moreover, the methodology proposed for such works are reversible and the ground conditions will be reinstated upon construction of works. Therefore, no impact to the SAI is anticipated.

3.10.1.5 As control measures, the contractor should submit a method statement for the installation and removal of temporary works within the Ngau Tam Mei SAI for agreement with AMO. The works must ensure that no disturbance to the soil matrix at the Ngau Tam Mei SAI. Photographic records of the ground condition should be taken before the execution of any works, after all works have been removed from the zone, as well as at intervals during the construction phase and submitted to AMO for retention.

3.10.1.6 According to the latest construction works arrangement, the works to be carried out within the Ngau Tam Mei SAI would only be those temporary works that are non-intrusive in nature and would cause no sub-soil disturbance. However, if the arrangement deviates from the current proposed situations, an archaeological survey-cum-excavation should be carried out to obtain archaeological data for assessing the impact before commencement of the works, in consultation with the AMO, and to design mitigation measure as appropriate.

3.10.1.7 Pursuant to the Antiquities and Monuments Ordinance (Cap. 53), the project proponent is required to inform the Antiquities and Monuments Office (AMO) immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

3.11 Hazard to Life

3.11.1.1 The hazard assessment has been conducted in accordance with the relevant requirements as specified in Clause 3.5.12 and Appendix K of the EIA Study Brief, as well as the criteria and guidelines as stated in Annex 4 of the EIAO-TM.

3.11.1.2 To ensure timely delivery to blasting site and maintain the construction process, a temporary explosive magazine site at Tai Shu Ha (Yuen Long) for overnight storage of explosives is required.

3.11.1.3 The magazine site at Tai Shu Ha (Yuen Long) would follow the relevant design requirements in terms of sufficient separation distances and the design of the storage facilities. Review suggests that the associated risk for the operation of the temporary magazine site would be insignificant.

3.11.1.4 The assessment results show that the individual risks associated with the storage, transport and use of explosives complied with the Risk Guidelines as stated in Annex 4 of the EIAO-TM.

3.11.1.5 The societal risks associated with the storage and use of explosives fall into the "Acceptable" region of the Risk Guidelines; while the societal risk associated with the transport of explosives falls into the lower "as low as reasonably practicable" (ALARP) region of the Risk Guidelines, mitigation measures for the transport of explosives were explored and assessed by using cost-benefit analysis (CBA). Justified mitigation measures and good practices were recommended.

4. ENVIRONMENTAL MONITORING AND AUDIT (EM&A)

4.1.1.1 The EIA Study of the Project has demonstrated its compliance with the EIAO-TM requirements. Actual impacts during the construction and operational phases of the Project will be monitored through a detailed EM&A programme. Full details of the programme are presented in a separate EM&A Manual associated with the EIA Report. The EM&A programme will provide management actions and detail the recommended mitigation measures to check the effectiveness of the recommended mitigation measures and compliance with relevant statutory criteria, thereby ensuring the environmental acceptability of the construction and operation of the Project.

5. SUMMARY OF ENVIRONMENTAL OUTCOMES

5.1.1.1 The EIA has provided an assessment of the potential environmental impacts associated with the construction and operation of the Project.

Table 5.1 Summary of Key Environmental Problems Avoided and Sensitive Areas Protected

Design Approaches	Environmental Issues Avoided/Minimised and Sensitive Areas Protected
Avoidance of ecologically sensitive areas	• The current alignment avoids encroaching ecologically sensitive areas such as LTCP, the proposed Nam Sang Wai Wetland Conservation Park and natural watercourses at Ngau Tam

Design Approaches	Environmental Issues Avoided/Minimised and Sensitive Areas Protected
	Mei, thereby avoiding potential ecological impact. • The current alignment avoids encroaching the existing bat roost and planned bat shelter at Pok Wai, thereby avoiding the direct ecological impact to bats. • Locations of piers also designed to avoid direct encroachment into natural watercourses.
Relocation of I/C-03 to South of Ngau Tam Mei to avoid wetland habitat near NTMWTW	• The location of I/C 03 has been relocated from near the existing NTMWTW to near the South of Ngau Tam Mei NDA such that the direct impact to the pond and marsh habitats near NTMWTW has been avoided.
Avoid Habitat Isolation and Fragmentation and Roadkill Using Intermittent At-grade and Viaduct Design	• Adopt viaduct design as much possible to avoid habitat fragmentation and isolation at LTCP • For at-grade sections, retaining wall on the slope would be constructed to avoid wildlife roadkill
Viaduct Level Arrangement and Carriageway Separation for Enhancing Sunlight Penetration and Ecological Benefits	• Viaduct structures crossing wetland habitats are designed with specific height and width ratios and adequate headroom to minimize shading effect and allow optimal sunlight penetration • Piers are designed to avoid direct encroachment onto natural streams
Adoption of non-dredged method for works at Kam Tin River	• Minimised the disturbance to the riverbed • Minimised the impact to the water quality
Synergistic Woodland Enhancement near She Leng	• Enhance the habitat connectivity and quality of She Leng through off-site woodland compensation
Minimization of slope cutting	• Minimised disruption to the landscape resources and characters • Minimised potential impacts to the natural habitats and associated ecology • Minimised the amount excavation spoil and generation of C&D materials
Avoidance of direct impacts on SAI	• Avoid any excavation works which will cause potential disturbance to the soil matrix within the boundary of Ngau Tam Mei SAI to minimize the potential archaeological impact.
Adopt D&B method for tunnel construction	• Adopt D&B method over TBM method to minimize the generation of C&D materials
Adoption of environmentally friendly construction methods and phased construction	• Adoption of off-site precast structures / modular integrated construction method to minimise the impacts on air quality, noise, water quality and waste during construction. • Adopt phased construction arrangement to minimise the cumulative environmental impacts. • Implementation of 100m buffer area from the existing Sha Po Ardeid Night Roost, where time control on the use of noisy construction activities would be implemented to minimize disturbance to the night roost.

Design Approaches	Environmental Issues Avoided/Minimised and Sensitive Areas Protected
Implementation of 'no work zone' for minimization of habitat loss	• Minimised the impacts and disturbance to ecologically sensitive areas and resources
Avoidance of illegal dumping	• The recommended preventive measures, such as real time tracing and monitoring of their travel routings and parking locations of the dump trucks through GPS or equivalent system, would avoid / minimise the chance of illegal dumping.
Implementation of Environmental Monitoring and Auditing System	• Ensure all the recommended measures are properly in place and their effectiveness.

6. CONCLUSION

6.1.1.1 The findings of the EIA provided information on the nature and extent of the environmental impacts likely to arise from the construction and operation of the Project. The EIA has, where appropriate, identified mitigation measures to ensure compliance with environmental legislations and standards. The summary of the environmental impacts arising from the Project is presented in **Table 6.1**.

6.1.1.2 Overall, the EIA concluded that the Project would comply with the requirements of the EIA Study Brief (ESB-373/2025) and the EIAO-TM with the implementation of the proposed mitigation measures during the construction and operation phases. An EM&A programme and the implementation schedule of the proposed mitigation measures have been provided and recommended in a stand-alone EM&A manual to check the effectiveness of the proposed mitigation measures.

Table 6.1 Summary of Environmental Impacts

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
Air Quality Impact					
Construction Impact					
Representative residential, commercial developments and government uses within 500m from the boundary of the Project Site.	• Potential air quality impact from the construction works of the Project would mainly be related to construction dust from site formation, site clearance, excavation, slope works, construction of administration buildings and satellite control buildings, blasting for tunnel construction, mucking out blasted rock and soil from drill-and-blast operations, material handling and backfilling, foundation works, handling and stockpiling of dusty materials, wind erosion of exposed area, as well as use of on-site powered mechanical equipment, and emissions from	• Annexes 4 and 12 of the EIAO-TM	N/A	Air quality control measures stipulated in Air Pollution Control (Construction Dust) Regulation and good site practices listed below should be carried out to further minimise construction air quality impact: • Regular watering to reduce dust emissions from exposed site surfaces and unpaved roads, particularly during dry weather; • Frequent watering for particularly dusty construction areas and areas close to ASRs; • For the works sites close to the ASRs with a separation distance less than 10 m as shown in Appendix 3.1, provide hoardings of not less than 3.5 m high from ground level along the site boundary; for the other works sites in general, provide hoarding of not less than 2.4 m	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	construction vehicles. • Construction activities of the concurrent projects within 500m assessment area would also pose cumulative air quality impact.			high from ground level along site boundary except for site entrance or exit; • Avoid positioning material stockpiling areas, major haul roads and dusty works within the construction site close to concerned ASRs; • Avoid unnecessary exposed earth; • Locate all the dusty activities away from any nearby ASRs as far as practicable; • Open stockpiles shall be avoided or covered. Where possible, prevent placing dusty material storage piles near ASRs; • Tarpaulin covering of all dusty vehicle loads transported to, from and between site locations; • Establishment and use of vehicle wheel and body washing facilities at the exit points of the site; • Where possible, routing of vehicles and positioning of construction plant should be at	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				the maximum possible distance from ASRs; • Imposition of speed controls for vehicles on site haul roads; and • Instigation of an EM&A programme with continuous construction dust monitoring to monitor the construction process in order to enforce controls and modify method of work if dusty conditions arise. Guidelines stipulated in EPD's Recommended Pollution Control Clauses for Construction Contracts should be incorporated in the contract document to abate construction air quality impacts. These clauses include: • The Contractor shall observe and comply with APCO and its subsidiary regulation, particularly the Air Pollution Control (Open Burning) Regulation, Air Pollution Control (Construction Dust) Regulation, Air Pollution Control (Non-road Mobile Machinery)(Emission) Regulation, Air Pollution Control	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				(Fuel Restriction) Regulation, and Air Pollution Control (Smoke) Regulation; • In addition to the statutory requirements of the Regulations, the Contractor of the public works contracts shall also observe the requirements as set out in the government circulars, including DEVB's TC No, 13/2020 (Timely Application of Temporary Electricity and Water Supply for Public Works Contracts and Wider Use of Electric Vehicles in Public Works Contracts) and DEVB's TC No. 1/2015 (Emissions Control of NRMM in Capital Works Contracts of Public Works); • The Contractor shall undertake at all times to prevent dust nuisance as a result of the construction activities, and minimise the emission of air pollutants from construction plant and equipment; • The Contractor shall ensure that there will be adequate water	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				supply / storage for dust suppression; • The Contractor shall devise and arrange methods of working and carrying out the works in such a manner so as to minimise dust impact on the surrounding environment, and shall provide experienced personnel with suitable training to ensure that these methods are implemented properly; • For better smoke control, the Contractor shall not use diesel hammer for percussive piling; and • Before the commencement of any work, the Engineer may require the methods of working, plant, equipment and air pollution control system to be used on the site to be made available for inspection and approval to ensure that they are suitable for the project. In order to help reduce pollution, timely application of temporary electricity and water supply would	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				be made and electric vehicles would be adopted in accordance with DEVB TC(W) No. 13/2020 – Timely Application of Temporary Electricity and Water Supply for Public Works Contracts and Wider Use of Electric Vehicles in Public Works Contracts in the Project. To minimise the exhaust emission from NRMMs during the construction phase, DEVB TC(W) No. 1/2015 - Emissions Control of NRMM in Capital Works Contracts of Public Works shall be followed, and the following measures should be applied: • Connect construction plant and equipment to main electricity supply and avoid use of diesel generators and diesel-powered equipment as far as practicable; • Avoid exempted NRMMs; and • Deploy electrified NRMMs as far as practicable. For the tunnelling works by drill-and-blast, a blast door should be installed at the opening to avoid the escape of fugitive dust from the	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				blasting. The blast door should remain closed and ventilation with dust filter of at least grade F8 in accordance with EN 779:2012 or grade E11 as per EN 1822 would be installed at the exhaust of the ventilation system, which is able to capture 90% to 95% of 0.4 μm particles. Water spaying should be applied to facilitate dust settlement. The areas within 30m from the blasting area should be wetted with water prior to blasting and blasting shall not be carried out when the strong wind signal or tropical cyclone warning signal No. 3 or higher is hoisted.	
Operation Impact					
Representative residential, commercial developments and government uses within 500m from the boundary of the Project site.	Air Quality Impact NO_2 • 19th highest 1-hr average conc.: 47.9 – 164.2 μg/m^3 • 10th highest 24-hr average conc.: 16.8 – 76.2 μg/m^3 • Annual average conc.: 8.2 – 37.2	Prevailing AQOs NO_2 • 1-hr average conc.: 200 μg/m^3 (Number of exceedances allowed: 18). • 24-hr average conc.: 120 μg/m^3 (Number of	N/A	• No adverse air quality impact is anticipated during the operational phase of the Project, thus mitigation measure is deemed not necessary.	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	μg/m^3 RSP • 10th highest 24-hr average conc.: 50.3 – 55.6 μg/m^3 • Annual average conc.: 19.5 – 21.5 μg/m^3 FSP • 19th highest 24-hr average conc.: 28.9 – 33.8 μg/m^3 • Annual average conc.: 12.0 – 13.5 μg/m^3	exceedances allowed: 9). • Annual average conc.: 40 μg/m^3. RSP • 24-hr average conc.: 75 μg/m^3 (Number of exceedances allowed: 9). • Annual average conc.: 30 μg/m^3. FSP • 24-hr average conc.: 37.5 μg/m^3 (Number of exceedances allowed: 18). • Annual average conc.: 15 μg/m^3.			
Noise Impact					
Construction Impact					
Representative domestic premises, educational institutions, places of public worship and	• Potential adverse construction noise impact to existing NSRs within 300 m from the boundary of the Project Site due to	• Annexes 5 and 13 of the EIAO TM • Leq(30 min) 75 dB(A) at all domestic premises,	N/A	• Good site practices to limit noise emissions at the source; • Use of QPME and quieter construction methods; • Use of temporary noise barriers, noise enclosure, noise insulating	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
noise sensitive temporary structures within 300 m from the boundary of the Project Site that rely on opened windows for ventilation.	construction works from the Project	temporary housing accommodation, hostels, convalescent homes and home for the aged • Leq(30 min) 70 dB(A) at places of public worship, courts of law, hospitals and medical clinics and educational institutions (including kindergartens and nurseries) (with criterion of 65 dB(A) during examinations)		fabric, silencer, etc. to screen noise from construction equipment; • Install acoustic tunnel door or enclosure at the tunnel portals for construction activities to be carried out inside tunnel; and • Alternative use of plant items within one work site, wherever practicable.	
Operation Impact (Road Traffic Noise)					
Representative existing and planned residential uses, educational institutions, places of public worship, noise sensitive	• Predicted overall (including existing and project roads) noise levels: up to 84 dB(A) • Predicted road traffic noise levels of Project roads: up to 79 dB(A)	• Annexes 5 and 13 of the EIAO TM • Calculation of Road Traffic Noise (1988) by Department of Transport, UK • EIAO-GN 12/2023 • L10(1 hour) 70	• Exceedance of the noise criteria by up to 14 dB(A)	• Provision of low noise road surfacing (LNRS) • Provision of reflective/ absorptive type vertical barrier and cantilevered noise barriers on concerned Project road sections • At-receiver acoustic window and/ or acoustic balcony	• No residual impacts anticipated

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
temporary structures and planned residential developments within 300 m from the boundary of the Project Site that rely on opened windows for ventilation.		dB(A) at 1 m from the façade of residential dwellings / noise sensitive temporary structures • L10(1 hour) 65 dB(A) at 1 m from the façade of educational institutions and places of public worship			
Operation Impact (Fixed Noise)					
Representative existing and planned residential uses, educational institutions, places of public worship, barracks, noise sensitive temporary structures and planned	Fixed Noise Sources Impact • Adverse fixed noise impact is not anticipated due to proposed fixed noise sources with good design and mitigation measures, and EM&A.	• Annexes 5 and 13 of the EIAO TM • Appropriate Acceptable Noise Level (ANL) -5 dB(A) as shown in Table 2 of Technical Memorandum for The Assessment of Noise from Places other than Domestic	N/A	EPD's "Good Practices on Ventilation System Noise Control" • Quieter equipment; • Silencer; • Barrier; and • Enclosure, etc.	No residual impacts anticipated

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
residential developments within 300 m from the boundary of the Project Site that rely on opened windows for ventilation.		Premises, Public Places or Construction Sites (IND-TM) or the prevailing background noise level			
Water Quality Impact					
Construction Impact					
Representative WSRs within 500m from the boundary of the Project Site	The potential sources of water quality impact associated with the construction works include: • Construction works in Kam Tin River; • Wastewater from general construction activities; • Construction site run-off; • Accidental spillage of chemicals; • Sewage from construction workforce;	• Annexes 6 and 14 of the EIAO-TM • Water Quality Objectives for Deep Bay WCZ • Technical Memorandum on Standards for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS)	N/A	• Construction of pier and viaduct structures across Kam Tin River would require installation of cofferdam and/or watertight pile casing to enclose the construction works. The construction should be conducted by phases. Cofferdams should be designed by the Contractor which fulfil the requirements in DSD Technical Circular No. 1/2017 "*Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System*".	• No residual impacts anticipated

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	• Construction works near watercourses; • Removal / diversion of watercourses; • Removal / filling of ponds; • Groundwater infiltration arising from tunnel boring; and • Groundwater from contaminated areas, contaminated site run-off and wastewater from land decontamination.	• Practical Note for Professional Persons (ProPECC) PN 2/24 • Environmental, Transport and Works Bureau (ETWB) Technical Circular (Works) No. 5/2005 • Chapter 9 of Hong Kong Planning Standards and Guidelines (HKPSG)		• Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdam and watertight pile casing to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharge. • Silt curtain should be deployed to completely enclose the cofferdam and watertight pile casing prior to conduct construction works in Kam Tin River, and installation and removal of cofferdam and watertight pile casing. • Mitigation measures and good site practices in ProPECCPN 2/24 "*Construction Site Drainage*" • Practices in ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*" • Waste Disposal Regulation (Cap. 354)	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				• Provision of interim treatment facilities, such as chemical toilets, for construction workforce	
Operation Impact					
Representative WSRs within 500m from the boundary of the Project Site	Potential water quality impacts associated with the operation phase include: • Water flow and water quality impacts on Kam Tin River; and • Non-point source surface runoff from new impervious areas. • Sewage effluent generated from the new administration building.	• Annexes 6 and 14 of the EIAO-TM • Water Quality Objectives for Deep Bay WCZ • Technical Memorandum on Standards for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS) • ProPECC PN 1/23	N/A	• Stormwater Management Practices and Stormwater Pollution Control Plan to reduce non-point source pollution. • Adequate design in silt trap for the new road drainage which take into account the guidelines in ProPECC PN 1/23 "*Drainage Plan subject to Comments by the EPD*". • Sewage from the new administration building would be collected and discharged to the public sewerage system and divert to sewage treatment works for further treatments.	• No residual impacts anticipated
Waste Management Implications					
Construction Impact					

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
N/A	• Around 5,495,370 m^3 of inert C&D materials would be generated during the construction phase, while 109,910 m^3 would be reused on-site and 5,385,470 m^3 would be delivered to Public Fill Reception Facility, subject to allocation by the Public Fill Committee and/or reused in other concurrent projects; • Around 174,220 m^3 of non-inert C&D materials would be generated; • Approximately 11,080 m^3 of excavated sediment would be generated during the construction of piling and pier; • Chemical waste in the order of few hundred litres per month would be generated;	• Annexes 7 and 15 of the EIAO-TM • Waste Disposal Ordinance (Cap. 354) • Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C) • Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N) • Land (Miscellaneous Provisions) Ordinance (Cap. 28) • Public Health and Municipal Services Ordinance (Cap. 132BK) – Public Cleansing and	N/A	• Implementation of good site practices, waste reduction measures and proper storage, collection and transportation of waste.	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	• Around 17,550 tonnes of general refuse would be generated.	Prevention of Nuisances Regulation • Dumping at Sea Ordinance (DASO) (Cap.466) • Project Administration Handbook for Civil Engineering Works (PAH)			
Operation Impact					
N/A	• Maximum of about few hundred litres per month of chemical waste would be generated during maintenance and servicing activities; • Maximum of about 408 kg per day of general refuse would be generated from the employees in the administration buildings.	• Annexes 7 and 15 of the EIAO-TM • Waste Disposal Ordinance (Cap. 354) • Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C) • Public Health and Municipal Services Ordinance (Cap.	N/A	• Implementation of good site practices, waste reduction measures and proper storage, collection and transportation of waste.	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
		132BK) – Public Cleansing and Prevention of Nuisances Regulation			
Land Contamination					
Onsite construction workers and future user	• Potential land contaminating land use was identified within the Project area.	• Annex 19 of the EIAO-TM • Guidance Note for Contaminated Land Assessment and Remediation (EPD, 2023) • Practice Guide for Investigation and Remediation of Contaminated Land (EPD, 2023) • Guidance Manual for Use of Risk-based Remediation Goals for Contaminated Land Management (EPD, 2023)	N/A	• Further land contamination assessments, including further site appraisal and submission of CAP(s), should be conducted for the whole Project Site at a later stage of the Project when site access is available to confirm the existing land uses / activities, identify the presence of any potential contamination sources, and address any new contamination issues. The associated SI works and any necessary remediation action are recommended to be carried out after operation of concerned site(s) has ceased but prior to the commencement of construction works. The recommended further assessment and remediation works, including the submission of CAP(s), CAR(s)/RAP(s) and RR(s) would follow the relevant Guidance Manual, Guidance	• No residual impact is anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				Note and Practice Guide	
Onsite construction workers and future user	• Natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories.	• EIAO-TM • Environmental Impact Assessment Report – North East New Territories New Development Areas (EIA Register No. AEIAR-175/2013)	Arsenic area (>571 mg/kg) as extracted from Hong Kong Environmental Database	• A separate arsenic assessment and remediation plan should be prepared, in consultation with EPD, and submitted to EPD for endorsement.	• No residual impact is anticipated.
Ecological Impact					
Construction Impact					
Ecological resources likely to be impacted by the Project	• Permanent loss of wetland habitats and watercourse, and wooded and vegetated habitat • Fragmentation and isolation of wooded areas and wetland habitats • Direct impact on floral and faunal species of conservation importance • Potential direct injury/	• Annex 8 and Annex 16 of EIAO-TM • EIAO Guidance Notes Nos. 3/2010, 6/2010, 7/2023 and 10/2023	N/A	• Avoid and/or minimise direct impacts on recognised sites of conservation importance and other ecologically sensitive resources • Careful planning during detailed detail stage to minimise works footprint • Minimize potential disturbance to the Sha Po ardeid night roost by establishing buffer area with timing control of construction activities (avoiding noisy works from one hour before sunset to one hour after sunrise)	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	mortality of wildlife species (e.g. bird collision) • Disturbance impact (e.g. noise, glare, air/dust, traffic, anthropogenic activities and water quality deterioration etc.) to recognised sites of conservation importance, ecologically sensitive resources, flight corridor, night roost, planned bat roost, natural habitats and associated wildlife in the vicinity • Potential barrier effect and impact on microclimate on the planned bat shelter at Pok Wai • Potential water quality and water flow impact • Groundwater drawdown from tunnel construction • Night-time disturbance • Ground-borne noise and vibration impact			• Adoption of non-dredged method for construction works at the KTMDC • Adoption of no-work zones • Piling works at the Kam Tin River would be scheduled to conduct during high tidal level (i.e. above 1.0 mCD) during dry season when the riverbed of the Kam Tin River would be mostly submerged. • On-site Compensation for the unavoidable loss of marsh/reed, ponds, natural watercourses, and modified watercourse • Off-site woodland compensation • Pre-construction survey and transplantation / translocation for the affected species of floral and faunal conservation importance • Avoid and/or minimize wildlife species from entering construction site by erecting hoarding or barrier to restrict and confine the construction activities within the Project Boundary, thereby preventing wildlife from accidentally entering the area.	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				• Minimize bird collision by using low-reflective materials (e.g. tinted glass, low-reflective window film). Non-transparent panels should also be used as noise enclosures, along with non-glaring tinted materials or dark patterns superimposed on the majority of glazing along noise barriers. • Avoid and/or minimise night-time disturbance through the use of directional lighting to prevent light spill into sensitive areas (e.g. the planned bat shelter, the potential recipient site of Motschulsky's Starworm in Ki Lun Tsuen, and recognised sites of conservation importance such as LTCP and CAs to the south of the Project Boundary, WBA, and WCA to the west of Tsing Long Highway, which are near construction site) and controlling of night-time lighting periods, particularly for works site(s) located in proximity to these areas. • To further minimise potential adverse impacts to the ecologically sensitive habitats (i.e. KTMDC and ponds near	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				NTMWTW), phasing of construction works would be adopted in these areas under the Project. This phased construction in KTMDC and ponds near NTMWTW aims to reduce the scale of disturbance and the total area of construction activities. • The area occupied by actual construction (i.e. pond habitat near NTMWTW and Kam Tin Main Drainage Channel (KTMDC)) would not exceed half of the designated temporary works area at any given time. The remaining half would be maintained as pond habitat or watercourse. The impacted areas from temporary works will be reinstated to their pre-existing condition as much as practicable before commencing construction on the remaining half. • Avoid and/or minimise potential noise, air and water quality impact through implementation of mitigation measures and good site practices • Provision of screening and use of directional lighting to control glare and light spill	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
Operation Impact					
Ecological resources likely to be impacted by the Project	• Potential direct impact on recognized sites of conservation importance and ecologically sensitive resources • Potential fragmentation and isolation of vegetated habitats • Water quality / water holding/water flow impacts • Potential disturbance impacts (e.g. air quality, noise, light pollution, traffic and visual) to the recognized sites of conservation importance, ecologically sensitive resources, species of conservation importance, wildlife, and habitats • Potential bird collision with man-made structures • Potential impacts from direct injury/mortality due to at-grade sections	• Annex 8 and Annex 16 of EIAO-TM • EIAO Guidance Notes Nos. 3/2010, 6/2010, 7/2023 and 10/2023	N/A	• No further direct impact on recognized sites of conservation importance and ecologically sensitive resources • At-grade permanent structures would primarily be located along the edges of wooded habitats, where significant wildlife crossings between disturbed area (e.g. village/orchard and developed area/wasteland habitats) and wooded habitats are not anticipated. In addition, part of the at-grade section within the vegetated habitats at Kai Kung Leng would take the form of a viaduct, while a substantial portion of the alignment in Ngau Tam Shan and the southeast of Hadden Hill is designed as underground tunnels. This design effectively minimises fragmentation and facilitates animal passage between the vegetated and wooded habitats. • The recognised sites of conservation importance, ecologically sensitive resources, species of	• No residual impacts anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	of the alignment • Potential obstruction and disturbance to flight corridor			conservation importance, wildlife, and habitats are already subject to substantial and ongoing human activities, including existing roads, high traffic volumes, nearby residential and industrial developments, and other infrastructure. As a result, the potential additional disturbance impact on these sites of conservation importance and ecologically sensitive resources during the operation phase is anticipated to be low. • An adequate drainage system would be implemented for the proposed at-grade road, viaduct, and tunnel which would effectively prevented surface runoff from entering ecologically sensitive watercourses. • Use of non-transparent or non-glaring materials on noise barriers • For the at-grade sections near LTCP and Long Ha, animal barrier and retaining wall on the slope would be constructed to avoid wildlife roadkill • The proposed viaduct section	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				would be of similar height and close to the existing viaduct of the Yuen Long Highway and Tsing Long Highway, avifauna species are expected to continue transversing beneath or above the viaduct sections, similar to current conditions. In terms of disturbance, the anticipated traffic flow on the proposed viaduct section would also be comparable to that of the existing viaducts in the KTMDC, where the surrounding environment is already subjected to constant human disturbance and noise from heavy traffic. In view of these analogous conditions, the potential obstruction and disturbance to the flight corridor and foraging ardeids and waterbirds in the KTMDC and the Sha Po Night Roost are anticipated to be low. • Directional lighting would be adopted to further minimise potential light spill to the planned bat shelter at Pok Wai.	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
Fisheries					
Fisheries resources and habitats likely to be impacted by the Project	• Negligible impacts from loss of fishponds • Deterioration of water quality and hydrological condition • Bund stability • Blockage of access	• Annex 9 and Annex 17 of EIAO-TM • Water Pollution Control Ordinance (Cap. 358)	N/A	• Maintaining bund stability • Minimisation of potential water quality impacts • Control of construction site run-off • Control of construction-related activities • Implementation of good site practices	• No residual impacts anticipated.
Landscape and Visual					
Construction Impact					
Existing LRs and LCAs within the assessment area	• Key affected LRs: • Negligible LR1.06, LR1.07, LR1.11, LR1.12, LR2.06, LR2.07, LR2.09, LR3.08, LR3.09 and LR3.10; • Slight LR2.03, LR2.04, LR2.05, LR2.10, LR3.03, LR3.06 and LR3.07; and • Moderate LR1.01, LR1.02,	• Annexes 10 and 18 of the EIAO-TM Environmental Impact Ordinance Guidance Note No. 8/2023	N/A	• CM1 - Trees without impact from the proposed works should be retained and any existing trees unavoidably affected by the works should be transplanted as far as possible in accordance with DEVB TC(W) No. 4/2020; • CM2 - Erection of decorative screen hoarding with decorative effect or planting compatible with the surrounding setting to minimise potential landscape impacts; • CM3 - Construction facilities and activities on work sites and areas should be carefully	• Key affected LRs: • Negligible LR1.06, LR1.07, LR1.11, LR1.12, LR2.06, LR2.07, LR2.09, LR3.08, LR3.09 and LR3.10; • Slight LR1.03, LR1.04, LR1.05, LR1.10, LR2.03, LR2.04, LR2.05, LR2.10, LR3.03, LR3.06 and LR3.07; and • Moderate LR1.01, LR1.02, LR2.01, LR2.02, LR2.11, LR2.12,

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	LR1.03, LR1.04, LR1.05, LR1.10, LR2.01, LR2.02, LR2.11, LR2.12, LR3.01, LR3.02, LR3.04, LR3.05 and LR3.12. • Key affected LCAs: • Negligible LCA1.7 and LCA 3.6; • Slight LCA1.5, LCA2.2, LCA2.4, LCA2.5, LCA2.6, LCA3.2, LCA3.5, LCA3.7 and LCA3.8; and • Moderate LCA1.2, LCA1.3, LCA1.8, LCA2.3, LCA3.1, LCA3.3 and LCA3.4.			managed and controlled, including height and disposition /arrangement, to minimise any potential adverse landscape impacts; • CM4 - Reinstatement of the temporarily affected landscaped area within assessment area in like-for-like basis would be implemented to restore the existing natural environment as far as possible; • CM5 - The optimum alignment of proposed road works is carefully designed to minimize direct conflict with the LTCP and Conservation Area. The works area of road works and associated slope works shall be minimized and confined to avoid any unnecessary vegetation loss in the adjacent Country Park and Conservation Area; and • CM6 - The design shall minimize disturbance on watercourses/wetlands, particularly for natural watercourse/wetlands. Good site practices as described in ETWB TCW No. 5/2005 “Protection of natural	LR3.01, LR3.02, LR3.04, LR3.05 and LR3.12. • Key affected LCAs: • Negligible LCA1.7 and LCA 3.6; • Slight LCA1.3, LCA1.5, LCA1.8, LCA2.2, LCA2.4, LCA2.5, LCA2.6, LCA3.2, LCA3.5, LCA3.7 and LCA3.8; and • Moderate LCA1.2, LCA2.3, LCA3.1, LCA3.3 and LCA3.4.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				streams/rivers from adverse impacts arising from construction works" shall also be adopted to avoid any pollution entering the watercourses nearby where applicable.	
Operation Impact					
Existing LRs and LCAs and Key Public Viewers / VPs within the assessment area	• Key affected LRs: • Negligible LR1.06, LR1.07, LR1.11, LR1.12, LR2.06, LR2.07, LR2.09, LR3.08, LR3.09 and LR3.10; • Slight LR2.03, LR2.04, LR2.05, LR2.10, LR3.03, LR3.06 and LR3.07; and • Moderate LR1.01, LR1.02, LR1.03, LR1.04, LR1.05, LR1.10, LR2.01, LR2.02, LR2.11, LR2.12, LR3.01, LR3.02, LR3.04, LR3.05 and LR3.12.	• Annexes 10 and 18 of the EIAO-TM • Environmental Impact Ordinance Guidance Note No. 8/2023		• OM1 - Aesthetically Pleasing Design of Aboveground Structures according to ACABAS submission upon completion of conceptual design should be in accordance with ETWB TCW No. 36/2004 while aesthetic treatment and Design of the noise mitigation measures, building and tunnel ventilation shaft should be vetted and advised upon by ArchSD in accordance with ETWB TCW No. 8/2005. • OM2 - Highways Structures such as viaducts proposed shall be sensitively designed in the regard of form, tonal colour and texture so as to minimise any potential adverse landscape and visual impact. The slope associated structures such as proposed	• Key affected LRs: • Negligible LR1.06, LR1.07, LR1.11, LR1.12, LR2.03, LR2.04, LR2.05, LR2.06, LR2.07, LR2.09, LR2.10, LR3.03, LR3.06, LR3.07, LR3.08, LR3.09 and LR3.10; and • Slight LR1.01, LR1.02, LR1.03, LR1.04, LR1.05, LR1.10, LR2.01, LR2.02, LR2.11, LR2.12, LR3.01, LR3.02, LR3.04, LR3.05, and LR3.12. • Key affected LCAs: • Negligible LCA1.3, LCA1.5, LCA1.7, LCA1.8, LCA2.2, LCA2.4, LCA2.5, LCA2.6, LCA3.2, LCA3.5, LCA 3.6, LCA3.7 and LCA3.8; and

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	• Key affected LCAs: • Negligible LCA1.7 and LCA 3.6; • Slight LCA1.5, LCA2.2, LCA2.4, LCA2.5, LCA2.6, LCA3.2, LCA3.5, LCA3.7 and LCA3.8; and • Moderate LCA1.2, LCA1.3, LCA1.8, LCA2.3, LCA3.1, LCA3.3 and LCA3.4. • Key affected VPs: • Negligible VP-N.9 and VP-S.2; • Slight VP-N.2, VP-N.5, VP-N.6, VP-N.7, VP-N.8, VP-S.1, and VP-S.4; and • Moderate VP-N.1, VP-N.3, VP-N.4, and VP-S.3.			retaining structures shall be sensitively designed to minizine the extend of structure and to break down the monotonous façade with use of natural material, unobtrusive colour or texture to blend in the retaining structures to the adjacent landscape and visual context; • OM3 - As far as practicable, compensatory tree planting requirement of 1:1 in terms of number in accordance with DEVB TC(W) No. 4/2020 for compensation of felled trees subject to agreement made with relevant departmental parties; • OM4 - Buffer Planting shall be provided at the perimeter of the Portal Areas to screen and soften the proposed Aboveground Structures. Combination of shrub and groundcover, with tree planting as far as practicable, will be planted to enhance the buffer and screening effect; • OM5 - Landscape treatments on slope or retaining structure in accordance with GEO Publication No. 1/2011 –	• Slight LCA1.2, LCA2.3, LCA3.1, LCA3.3 and LCA3.4. • Key affected VPs: • Negligible VP-N.9 and VP-S.2; • Slight VP-N.5, VP-N.6 and VP-N.7; and • Moderate VP-N.1, VP-N.2, VP-N.3, VP-N.4, VP-N.8, VP-S.1, VP-S.3 and VP-S.4.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				Technical Guidelines on Landscape Treatment for Slopes shall be adopted subject to agreement made with relevant departmental parties; Woodland Mix Planting with tree whips shall be proposed as far as practicable in accordance with technical guidelines set out in GEO Publication No. 1/2011 at the tunnel portals and the affected slopes along the road improvement works. • OM6 - Post-planting monitoring of the compensatory trees shall be undertaken (namely, duration of the post-planting monitoring and monitoring methodology); • OM7 - In case any existing open space or sitting out area will be impacted upon during construction stage, all affected area will be reinstated and / or upgraded to integrate with the existing site context; • OM8 - Roof Greening should be proposed to the roof area of the proposed building works as far as practical to enhance the landscape quality of the structures and mitigate any potential visual impact on	

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				adjacent VSRs. All new building works are required to achieve the minimum standards and requirements on Site Coverage of Greenery in Table 1 of DEVB TC(W) No. 3/2012 – Site Coverage of Greenery for Government Building Projects. • OM9 - The proposed noise barriers & enclosures shall be design in an elegant manner that includes suitable combination of transparent and sound absorbent materials, appropriate colour selection of panels and supporting structures, or provision of at-grade planting of trees, shrubs and/or climbers. Aesthetic treatment and Design of the noise mitigation measures should be vetted and advised upon by ACABAS in accordance with ETWB TCW No. 36/2004	
Impact on Cultural Heritage					
Construction Impact					
Built Heritage	No adverse impact on built heritage is anticipated during the construction phase.	• EIAO-TM Annexes 10 and 19 • A&MO (Cap.53)	N/A	• No mitigation measure is required for built heritage.	• No residual impact anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
		• Requirement set out under Clause 3.5.11 and Appendix J of the EIA Study Brief			
Archaeological Resources	• While there might be temporary works conducting within the Project Boundary but outside the alignment, these temporary works would not cause soil disturbance during the construction phase. Therefore, no impact to the Ngau Tam Mei SAI is anticipated.	• EIAO-TM Annexes 10 and 19 • A&MO (Cap.53) • Requirement set out under Clause 3.5.11 and Appendix J of the EIA Study Brief.	N/A	• Ngau Tam Mei SAI is partially located within the Project Boundary but outside the proposed alignment. According to the latest design of the construction works arrangement, no sub-soil disturbance works will be carried out within the SAI. Therefore, no impact to the SAI is anticipated. • If the work arrangement deviates from the current proposed situations, an archaeological survey-cum-excavation should be carried out to obtain archaeological data for assessing the impact before commencement of the works, in consultation with the AMO, and to design mitigation measure as appropriate. • Pursuant to the Antiquities and Monuments Ordinance (Cap. 53), the project proponent is	• No residual impact anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
				required to inform the Antiquities and Monuments Office (AMO) immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.	
Operation Impact					
Built Heritage	• No adverse impact on built heritage is anticipated during the operation phase.	• EIAO-TM Annexes 10 and 19 • A&MO (Cap.53) • Requirement set out under Clause 3.5.11 and Appendix J of the EIA Study Brief	N/A	• No mitigation measure is required for built heritage.	• No residual impact anticipated.
Archaeological Resources	• No adverse impact on archaeology is anticipated during the operation phase.	• EIAO-TM Annexes 10 and 19 • A&MO (Cap.53) • Requirement set out under Clause 3.5.11 and Appendix J of the EIA Study	N/A	• No mitigation measure is required for archaeology.	• No residual impact anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
		Brief			
Hazard to Life					
• Population within 100m influence zone along portals, explosives storage in magazine site and the transportation routes • Sensitive receivers within 100m influence zone along the drill-and-blast tunnel, such as building, slope and boulder	*Individual risk* • Within criteria of 1E-05 per year *Societal risk* • Cumulative risk for the storage and use of explosives falls into the "Acceptable" region • Cumulative risk for the transport of explosives falls into ALARP region *Cost benefit analysis* • Options such as use of alternative method of construction, use of magazines closer to the construction sites, and use of smaller quantities of explosives considered are not	• Annex 4 of the EIAO-TM • Dangerous Goods Ordinance (Cap. 295)	N/A	• Implement the good practices to minimize the potential risks even further and ensure that the services, utilities, slopes, retaining walls, buildings and structures will not be affected through ground vibrations or other effects. • Blasting Works: The good practice could make reference to the latest guideline including, but not limited to, Practice Note for Authorized Persons and Registered Structural Engineers (APP-72) by Buildings Department (BD). • Overnight Storage of Explosives: The good practice could made reference to the latest guideline including, but not limited to, "Guidance Note No. GN 8 How to Apply for a Mode A Store Licence for	• No residual impact anticipated.

Sensitive Receivers / Assessment Points	Impact Prediction Results (Without Mitigation)	Key Relevant Standards/Criteria	Extents of Exceedance (Without Mitigation)	Impact Avoidance Measures / Mitigation Measures	Residual Impacts (After Implementation of Mitigation Measures)
	practicable after comparing the implementation cost with maximum justifiable expenditure			Storage of S1DG (Blasting Explosives)" by CEDD. • Transportation of Explosives: The good practice could made reference to the latest guideline including, but not limited to "Guidance Note No. GN 2 Approval of an Explosives Delivery Vehicle" and "Guidance Note No. GN 3 Application and Handling of a Conveyance Permit" by CEDD.	